Clive Rafiq

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Education

Arizona State University — Tempe, AZ
B.S.E. in Computer Systems Engineering — *Expected May 2028*

- **Coursework:** OOP & Data Structures, Computer Organization & Assembly, C/C++ Programming, Discrete Math

Experience

Backend Developer Intern — May 2026 – Aug 2026
World Salon — *New York, NY*

- Owned per-recipient read/unread tracking end to end on a serverless notification service (FastAPI, AWS Lambda, DynamoDB), designing the per-user state model, PATCH endpoint, and query filter
- Identified that Socket.io's stateful connections couldn't run on Lambda and built real-time delivery on API Gateway WebSockets instead, provisioned with Terraform
- Unblocked production twice by root-causing an ASGI adapter that failed every request under Python 3.12 and JWKS auth fetches breaking across container boundaries, using CloudWatch and container-level debugging
- Caught a race condition that marked in-flight notifications as successful by building a Bash harness (publish to SQS, poll DynamoDB to terminal state); reused it to verify idempotency and malformed-event isolation

Software Engineer Intern, Backend — Dec 2025 – Jan 2026
BFI Finance Indonesia — *Jakarta, Indonesia*

- Built a Go media-caching service for WhatsApp customer documents, replacing raw photo and video storage with cached URL references to extend retention from a 2-week limit to indefinite
- Wired up event-driven processing with NATS, object storage, and internal APIs to cache and serve images and video at scale
- Shipped Go microservices with Docker and Kubernetes into a production loan workflow used daily by 6,000 loan officers serving roughly 400K customers

Software Engineer Intern, Full-Stack — Jul 2024 – Jan 2025
PhysioActive Indonesia — *Jakarta, Indonesia*

- Shipped a web dashboard surfacing customer messages and performance metrics, cutting employee response time 77%
- Built Python ETL pipelines in BigQuery using LLMs to classify 10,000+ messages for same-day sentiment analysis

Leadership

Co-Founder | *TechTrek Indonesia* — Aug 2023 – Feb 2025

- Founded a STEM education nonprofit in Jakarta, running Python and VEX Robotics workshops for 30 local students

Projects

SlackDB | *FastAPI, Slack Bolt, Docker, PostgreSQL* | **AutoDB Track Winner**, VillageHacks ASU — Apr 2026

- Co-built a Slack bot bringing text-to-SQL, migration risk analysis, and query optimization into team channels
- Designed guardrails for LLM-generated SQL: a multi-stage approval flow with risk-scored migrations, sandbox validation, rollback detection, and approver enforcement to block destructive production changes
- Shipped the FastAPI backend (async HTTPX, Slack Bolt, interactive Block Kit) via Docker on Railway with a full audit log

Drift | *React, MediaPipe, WebSerial, ESP32/C++* | *StarkHacks (Purdue)* — Apr 2026

- Built a passive focus tracker fusing typing telemetry (WPM, pause and backspace rate), MediaPipe gaze and blink detection, and ambient noise/light into weighted Focus and Fatigue scores, calibrated against a 30-second per-user baseline
- Wrote a bidirectional WebSerial protocol and ESP32 firmware in C++, learned during the hackathon, streaming sensor readings to the browser and rendering the computed focus score back to an onboard OLED

Rician Denoising with Deep Learning for MRI Tumor Segmentation | *Python, FFDNet (PyTorch)* — Aug – Dec 2025

- Adapted FFDNet to remove Rician noise from roughly 3,000 MRI slices, improving PSNR by ∼11 dB and cutting mean squared error from ∼851 to ∼81
- Built the training and evaluation pipeline: preprocessing, noise-level conditioning, and per-class benchmarking

Technical Skills

Languages: Python, Go, Java, C++, JavaScript, SQL, Bash
Frameworks & Infra: FastAPI, Flask, React, NATS, Docker, Kubernetes, Terraform, AWS (Lambda, SQS, DynamoDB, API Gateway, CloudWatch), GitHub, Git, CI/CD (GitHub Actions), Claude
Data & Storage: PostgreSQL, Redis, MongoDB, BigQuery, Pandas, NumPy